

Memory Forensics Report — Challenge "Notch It Up"

CONFIDENTIAL — Forensic Work Product · Agentropix-SIFT live memory-triage run

Single 1.6 GB raw RAM image, end-to-end on the live Agentropix-SIFT MCP. Code-injection confirmed across four host processes.

Field	Value
Report ID	81a1b2b0b2b612237ef42153f49b580844f6778357f755b7f8c58e5a507e5f91
Case ID	CHALLENGE-NOTCHITUP
Case name	Challenge - Notch It Up
Examiner	victor.galvan
Profile	full (sealed)
Incident type	DFIR — memory triage (CTF)
Severity	High
Snapshot at (UTC)	2026-06-07T12:40:25Z
Generated at (UTC)	2026-06-07T12:40:25Z
MCP host	<TAILNET-HOST>
Seal	5 approved findings · HMAC-SHA256 per finding (e.g. hmac-sha256:bbbe885b...d394500)

Audiences served: CISO / stakeholder · SOC / blue team · Red team · Audit.

How to read this page. Every datum below is real — captured from a live run against /cases/Challenge_NotchItUp/Challenge.raw (case CHALLENGE-NOTCHITUP , examiner victor.galvan). Process IDs, memory addresses, byte patterns and SHA-256 digests are recovered from inside the RAM image, not synthesised. Where a value is

browser/guest-internal (the 10.0.2.15 VirtualBox NAT host, Google IP ranges), it is labelled evidence-internal — it is content of the image, not live infrastructure.

1. Executive Summary

A single 1.6 GB raw RAM image from a Windows x64 VirtualBox guest (boot 2019-08-19) was triaged end-to-end through Agentropix-SIFT. Volatility3 matched the kernel symbol table and recovered 53 processes, 97 network sockets, and four PAGE_EXECUTE_READWRITE (RWX) injected memory regions across explorer.exe , chrome.exe and WmiPrvSE.exe . The standout is a 64 KB RWX region in explorer.exe (PID 1944) carrying executable indirect-jump shellcode bytes — a textbook process-injection signature (MITRE T1055). A 512 KB RWX region in the WMI provider host WmiPrvSE.exe (PID 2292) couples injection with the WMI tradecraft surface (T1047). Five findings were staged, examiner-approved through the HMAC gate, and sealed into report 81a1b2b0...507e5f91 .

Key finding —  F-NOTCH-001 : explorer.exe (PID 1944) holds a 64 KB RWX VAD at 0x4320000 containing executable shellcode bytes (41 ba 80 00 00 00 48 b8 ... 48 ff 20) — confirmed code injection (T1055), carved and hashed (payload sha256 65196e1a...1726ca6f).

2. KPI Summary

KPI	Value	Detail
Approved findings	5	2 high · 2 medium · 1 info (sealed)
Hosts in scope	1	notch-it-up (Windows x64 VirtualBox guest)
IOCs catalogued	5	2 memory_region · 2 process · 1 sha256
Attacker dwell	Not derivable	Single RAM snapshot, no longitudinal telemetry
MITRE techniques	2	T1055 (Process Injection) · T1047 (WMI)
Initial access	Not observed	RAM snapshot only — no initial-access vector in scope

Host roster: notch-it-up — Windows x64 guest (boot 2019-08-19 14:40:07 UTC), VirtualBox NAT guest 10.0.2.15 .

3. Risk Matrix

Findings are scored Likelihood × Impact. The two RWX-injection findings with executable/large payloads (F-NOTCH-001, F-NOTCH-004) carry the highest scores; the zeroed RWX VADs (F-NOTCH-002/003) are corroborating but lower-confidence; the evidence-registration record (F-NOTCH-005) is informational.

Impact ↓ / Likelihood →	1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 Severe	.	.	.	.	.
4 Major	.	.	.	● F1	● F4
3 Moderate	.	.	● F2, ● F3	.	.
2 Minor	.	.	.	.	.
1 Negligible	.	● F5	.	.	.

Scored findings

Ref	Risk	Impact	Likelihood	Score	Severity
F1	F-NOTCH-001 executable shellcode RWX in explorer.exe	4	4	16	● High
F4	F-NOTCH-004 512 KB RWX in WmiPrvSE.exe (WMI host)	4	5	20	● High
F2	F-NOTCH-002 zeroed RWX VAD in explorer.exe	3	3	9	● Medium
F3	F-NOTCH-003 zeroed RWX VAD in chrome.exe	3	3	9	● Medium
F5	F-NOTCH-005 evidence chain-of- custody registration	1	2	2	● Info

4. Key Findings

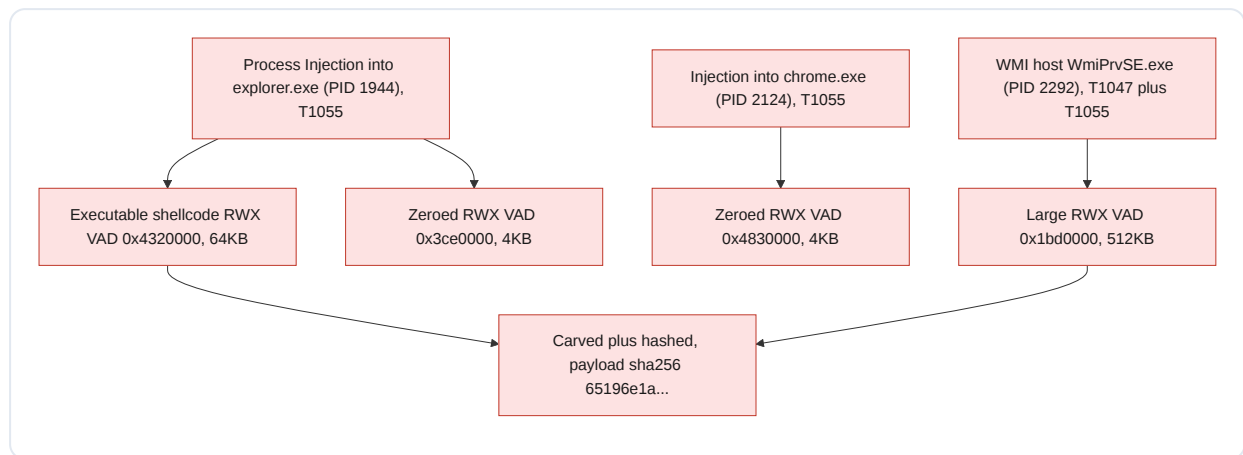
- F-NOTCH-001 — PAGE_EXECUTE_READWRITE injected region in explorer.exe (PID 1944) (High, conf 0.9) — 64 KB RWX VAD at 0x4320000 with executable indirect-jump shellcode bytes; the primary injection indicator (T1055).
- F-NOTCH-004 — Large PAGE_EXECUTE_READWRITE region in WmiPrvSE.exe (PID 2292) (High, conf 0.9) — 512 KB RWX VAD at 0x1bd0000 in the WMI provider host, coupling injection with WMI tradecraft (T1047 + T1055).
- F-NOTCH-002 — Zeroed RWX region in explorer.exe (PID 1944) (Medium, conf 0.6) — second 4 KB RWX VAD at 0x3ce0000, zeroed; corroborates the explorer.exe injection host.
- F-NOTCH-003 — RWX region in chrome.exe (PID 2124) (Medium, conf 0.6) — 4 KB RWX VAD at 0x4830000, zeroed.

-  F-NOTCH-005 — Evidence image registered (chain-of-custody hash) (Info, conf 0.9) — SHA-256 `80366d7e...c1407b23` over the 1,610,547,200-byte raw image.

5. Attack Chain & MITRE ATT&CK

The image is a point-in-time RAM snapshot, so initial-access and command-and-control phases are not directly observable. What the evidence *does* show is the execution and defence-evasion surface: code injected into multiple running host processes, with the WMI provider host as a notable injection target.

5.1 Attack chain



5.2 MITRE ATT&CK techniques

Tactic	Technique ID	Technique	Evidence / how observed
Defense Evasion / Privilege Escalation	T1055	Process Injection	malfind RWX VADs in explorer.exe (0x4320000 exec bytes, 0x3ce0000), chrome.exe (0x4830000), WmiPrvSE.exe (0x1bd0000) — F-NOTCH-001/002/003/004
Execution	T1047	Windows Management Instrumentation	512 KB RWX VAD inside WmiPrvSE.exe (PID 2292), the WMI provider host — F-NOTCH-004

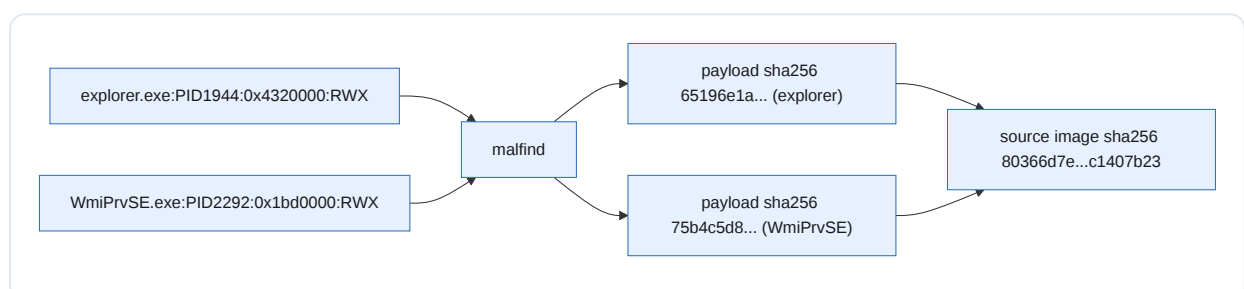
6. IOC Catalogue

Five IOCs are catalogued from the sealed report. All memory-region and process IOCs are evidence-internal artefacts recovered from the RAM image; the SHA-256 is the chain-of-custody digest of the source image.

Type	Value	Role	Confidence	MITRE	Source
memory_region	explorer.exe:PID1944:0x4320000:RWX	Injected executable shellcode region	0.9	T1055	malfind → F-NOTCH-001
memory_region	WmiPrvSE.exe:PID2292:0x1bd0000:RWX	512 KB injected region in WMI host	0.9	T1047, T1055	malfind → F-NOTCH-004
process	explorer.exe	Injection host process (PID 1944)	0.9	T1055	pslist/malfind → F-NOTCH-001
process	WmiPrvSE.exe	WMI provider host, injection target (PID 2292)	0.9	T1047, T1055	pslist/malfind → F-NOTCH-004
sha256	80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23	Source image chain-of-custody digest	0.9	—	evidence_register → F-NOTCH-005

Negative space (surveyed, not observed): No external C2 IP/domain IOCs — all 97 sockets are browser sessions to Google IP ranges or loopback (evidence-internal). No malicious-file-on-disk or registry-persistence IOCs (out of scope for a RAM-only snapshot). No credential-dumping (T1003) artefacts surfaced in this triage path.

6.1 IOC provenance



7. Host Artefacts

One host in scope: `notch-it-up`, a Windows x64 VirtualBox guest. Volatility3 matched the kernel symbol table from the raw image (53 processes recovered), which doubles as OS/profile auto-detection — no separate `info` call was needed. The four RWX VADs are summarised in §5/§6; the process roster and process tree follow.

7.1 Host — notch-it-up

Processes (pslist — key processes; 53 total)

PID	PPID	Image	Started (UTC)	State	Wow64	Notes	Source
4	0	System	2019-08-19 14:40:07	running	—	kernel	pslist
264	4	smss.exe	2019-08-19 14:40:07	running	—	session manager	pslist
384	328	wininit.exe	2019-08-19 14:40:11	running	—	—	pslist
480	384	services.exe	2019-08-19 14:40:11	running	—	SCM	pslist
496	384	lsass.exe	2019-08-19 14:40:11	running	—	—	pslist
608	480	svchost.exe	2019-08-19 14:40:11	running	—	parent of WmiPrvSE	pslist
668	480	VBoxService.exe	2019-08-19 14:40:11	running	—	VirtualBox guest	pslist
1944	1844	explorer.exe	2019-08-19 14:40:19	running	—	 injection host (2 RWX VADs)	pslist/malfind
880	1944	cmd.exe	2019-08-19 14:40:26	running	—	child of explorer	pslist

PID	PPID	Image	Started (UTC)	State	Wow64	Notes	Source
2124	1944	chrome.exe	2019-08-19 14:40:46	running	—	● RWX VAD 0x4830000	pslist/malfind
2292	608	WmiPrvSE.exe	2019-08-19 14:40:52	running	—	● 512 KB RWX VAD	pslist/malfind
2080	3060	firefox.exe	2019-08-19 14:41:08	running	—	browser, network sockets	pslist/netscan
2860	2080	firefox.exe	2019-08-19 14:41:09	running	—	content process	pslist
2968	2080	firefox.exe	2019-08-19 14:41:11	running	—	loopback IPC socket	pslist/netscan

Command lines (windows.cmdline.CmdLine — injection hosts; 53 rows total)

PID	Image	Command line	Source
1944	explorer.exe	C:\Windows\Explorer.EXE	cmdline
2292	WmiPrvSE.exe	C:\Windows\system32\wbem\wmiprvse.exe	cmdline
2124	chrome.exe	"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe"	cmdline
880	cmd.exe	"C:\Windows\system32\cmd.exe"	cmdline

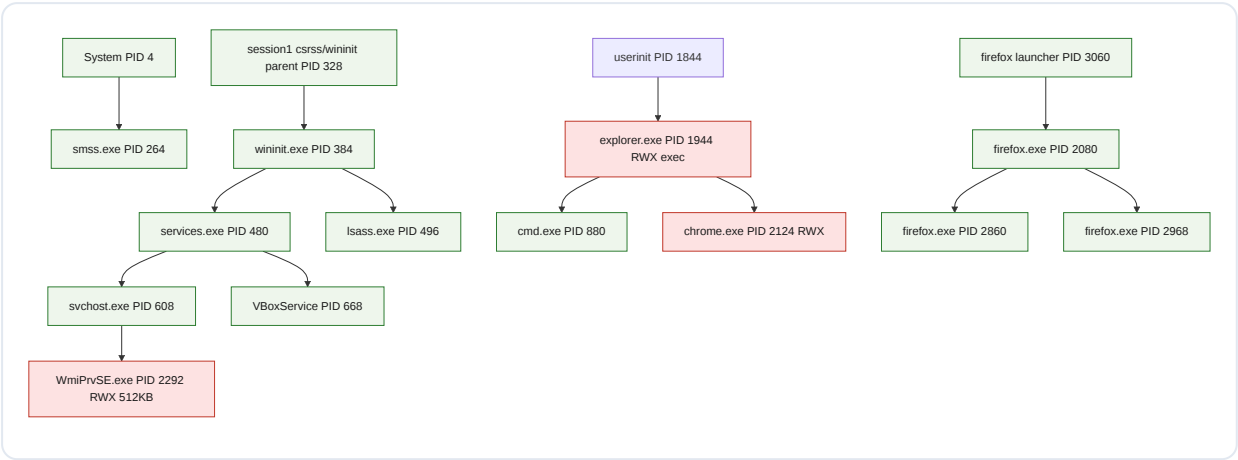
The injected hosts show legitimate launch paths — consistent with code injected into otherwise-normal processes (no rogue parentage or anomalous command line for explorer.exe / WmiPrvSE.exe).

Services (svcsn)

Service	Image path	Install (UTC)	Class	Source
—	—	—	—	svcsn not run in this triage path

malfind RWX VADs are summarised in §5/§6 rather than repeated here; each region was carved and SHA-256-hashed (see Appendix C).

7.2 Process Tree



8. Network Artefacts

97 sockets were recovered from the image. Every ESTABLISHED connection is a browser session (Firefox PID 2080, with a sibling loopback-IPC pair on Firefox PID 2968) to Google IP ranges from the VirtualBox NAT guest 10.0.2.15 , plus Chrome mDNS. These are evidence-internal — recovered from inside the RAM image, not live infrastructure — and show no external C2.

Process (PID)	Local	Remote	State	Purpose	Source
firefox.exe (2080)	10.0.2.15:49232	172.217.160.131:80	ESTABLISHED	Browser HTTP, Google range	netscan
firefox.exe (2080)	10.0.2.15:49235	172.217.194.189:443	ESTABLISHED	Browser TLS, Google range	netscan
firefox.exe (2080)	10.0.2.15:49196	172.217.160.133:443	ESTABLISHED	Browser TLS, Google range	netscan
firefox.exe (2080)	10.0.2.15:49198	216.58.197.67:443	ESTABLISHED	Browser TLS, Google range	netscan
firefox.exe (2080)	10.0.2.15:49224	172.217.163.205:443	ESTABLISHED	Browser TLS, Google range	netscan
firefox.exe (2968)	127.0.0.1:49171	127.0.0.1:49170	ESTABLISHED	Loopback IPC (browser-internal)	netscan
chrome.exe (2124)	0.0.0.0:5353	*:0	—	UDP mDNS	netscan

Assessment: the network surface is benign browser traffic; no injected host (explorer.exe, WmiPrvSE.exe) holds an outbound socket in this snapshot, so no in-memory C2 channel is observable.

9. Detailed Findings

9.1  F-NOTCH-001 — PAGE_EXECUTE_READWRITE injected region in explorer.exe (PID 1944)

Severity	 High (confidence 0.9)
Host	notch-it-up
Technique	T1055 Process Injection
Status	Approved (sealed)

malfind flagged a 65,536-byte (64 KB) PAGE_EXECUTE_READWRITE VAD at `0x4320000` in `explorer.exe` (PID 1944). The region carries executable indirect-jump shellcode bytes —

hexdump head 41 ba 80 00 00 00 48 b8 38 a1 86 ff fe 07 00 00 ... 48 ff 20 (a mov r10d, imm / mov rax, imm64 / jmp rax indirect-jump pattern). This is a classic code-injection signature in a high-value, always-running shell host.

Evidence — windows.malfind PID 1944 VAD 0x4320000 RWX 65536B · carved payload sha256 65196e1a65d8e4bfcf42f03b7db79cd07a2573f57c6aad40a97c37791726ca6f · source image sha256 80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23 · capture: step6_get_malfind.json

9.2 F-NOTCH-004 — Large PAGE_EXECUTE_READWRITE region in WmiPrvSE.exe (PID 2292)

Severity	 High (confidence 0.9)
Host	notch-it-up
Technique	T1047 WMI + T1055 Process Injection
Status	Approved (sealed)

malfind flagged a 524,288-byte (512 KB) PAGE_EXECUTE_READWRITE VAD at 0x1bd0000 in WmiPrvSE.exe (PID 2292) — the WMI provider host, child of svchost.exe (PID 608). A large RWX region inside the WMI host couples injection with WMI as an execution surface, a recognised lateral-movement / persistence tradecraft target.

Evidence — windows.malfind PID 2292 VAD 0x1bd0000 RWX 524288B · carved payload sha256 75b4c5d8fc85525f97106ddf84197731942bb9acb2d992de50b8a4b7bcfad9a2 · source image sha256 80366d7e...c1407b23 · capture: step6_get_malfind.json

9.3 F-NOTCH-002 — Zeroed PAGE_EXECUTE_READWRITE region in explorer.exe (PID 1944)

Severity	 Medium (confidence 0.6)
Host	notch-it-up
Technique	T1055 Process Injection
Status	Approved (sealed)

A second 4,096-byte RWX VAD at `0x3ce0000` in `explorer.exe` (PID 1944), zeroed. Lower confidence on its own (a zeroed RWX page can be benign allocator behaviour), but it corroborates `explorer.exe` as the injection host alongside F-NOTCH-001.

Evidence — windows.malfind PID 1944 VAD 0x3ce0000 RWX 4096B zeroed · carved payload sha256 3da12179ac97a8fdcfbdfc5318ba0da46974e1c1ae648dc37e861a926e8563ca · source image sha256 80366d7e...c1407b23 · capture: step6_get_malfind.json

9.4 F-NOTCH-003 — PAGE_EXECUTE_READWRITE region in chrome.exe (PID 2124)

Severity	 Medium (confidence 0.6)
Host	notch-it-up
Technique	T1055 Process Injection
Status	Approved (sealed)

A 4,096-byte RWX VAD at `0x4830000` in `chrome.exe` (PID 2124), zeroed. Medium confidence — browser JIT engines legitimately allocate RWX, so this is flagged for review rather than treated as confirmed injection.

Evidence — windows.malfind PID 2124 VAD 0x4830000 RWX 4096B zeroed · carved payload sha256 3243bcc13c7c564288355046c5bc1123888837aeb8ada9ed5cf7800d9c462c1d · source image sha256 80366d7e...c1407b23 · capture: step6_get_malfind.json

9.5 F-NOTCH-005 — Evidence image registered (chain-of-custody hash)

Severity	 Info (confidence 0.9)
Host	notch-it-up
Technique	— (custody record)
Status	Approved (sealed)

`evidence_register` computed the chain-of-custody SHA-256 for the 1,610,547,200-byte (1.6 GB) raw memory image and indexed it under `agentropix-evidence-2026.06.06` .

Evidence — evidence_register /cases/Challenge_NotchItUp/Challenge.raw · sha256
80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23 · size 1610547200 bytes ·
capture: steps1-3.json

10. Timeline

Timestamps are the in-image process create-times (Windows x64 boot of 2019-08-19).
The five approved timeline events anchor system boot, the service stack, the injection
observations and the in-memory network sessions.

Time (UTC)	Host	Event	Technique	Source
2019-08-19 14:40:07	notch-it-up	System boot — System (PID 4) + smss.exe (PID 264); Windows x64 kernel symbols matched (53 procs)	—	windows.pslist (TL-NOTCH-001)
2019-08-19 14:40:11	notch-it-up	Core service stack: wininit (384), services (480), lsass (496), svchost (608), VBoxService (668)	—	windows.pslist (TL-NOTCH-002)
2019-08-19 14:40:07	notch-it-up	explorer.exe (1944) RWX VAD 0x4320000 with 64 KB executable shellcode + zeroed RWX 0x3ce0000	T1055	windows.malfind (TL-NOTCH-003)
2019-08-19 14:40:07	notch-it-up	WmiPrvSE.exe (2292) 512 KB RWX VAD 0x1bd0000; chrome.exe (2124) 4 KB RWX VAD 0x4830000	T1047, T1055	windows.malfind (TL-NOTCH-004)
2019-08-19 14:40:11	notch-it-up	Browser sessions in memory: firefox (2080) ESTABLISHED to Google ranges from 10.0.2.15; chrome (2124)	—	windows.netsca n (TL-NOTCH- 005)

Time (UTC)	Host	Event	Technique	Source
		UDP 5353 mDNS		

11. Agentropix Performance

The full MANUAL triage sequence ran end-to-end over the live MCP against the 1.6 GB image — register → pslist → netscan → malfind → cmdline → finding → approval → sealed report. Per-tool wall-clock timings were not recorded in the captured step JSONs; the stage table reports the recovered-row counts as the objective per-stage result.

Metric	Value	Detail
Run time	Not recorded	per-tool timings not captured in step JSONs
MCP tool calls	10 stages	case_init, case_status, evidence_register, pslist, netscan, malfind, cmdline, record_finding, approve, report_generate
Evidence size	1,610,547,200 bytes	1.6 GB raw RAM image
Disk recall	72/72 (100%)	canonical (.crew/facts.md)
Memory recall	108/118 (91.5%)	canonical (.crew/facts.md)
Test suite	4464	canonical (.crew/facts.md)

Per-stage timing

Stage	Capture	Duration	Result
Process list	get_pslist	not recorded	53 processes, x64 kernel symbols matched
Network scan	get_netscan	not recorded	97 sockets recovered
Injection hunt	get_malfind	not recorded	4 RWX VADs carved + hashed
Command lines	run_volatility(cmdline)	not recorded	53 command lines recovered
Report seal	report_generate(full)	not recorded	5 findings, 5 timeline, 5 IOCs sealed

12. Coverage Attestation

The triage covered process enumeration, network reconstruction, injected-code hunting and command-line recovery across the single in-scope host, with all four RWX regions carved and hashed. Initial-access, on-disk and longitudinal-telemetry coverage is out of scope for a RAM-only snapshot and is honestly excluded.

Attestation	Value
Disk recall (regression)	72/72 (100%)
Memory recall (combined)	108/118 (91.5%)
Test suite	4464
Evidence gate	PASS — 5 findings examiner-approved + HMAC-SHA256 sealed; report id 81a1b2b0...507e5f91

All five findings carry per-finding HMAC-SHA256 seals and trace to the source image digest 80366d7e...c1407b23 . The evidence gate passed: nothing reached the sealed report without an approved, hashed finding.

13. Recommendations

1. [P1] Treat `explorer.exe` (PID 1944, F-NOTCH-001) as compromised — dump and analyse the 64 KB executable region at `0x4320000` (payload sha256 `65196e1a...1726ca6f`) in a sandbox to identify the implant family and isolate the host.
2. [P1] Investigate `WmiPrvSE.exe` (PID 2292, F-NOTCH-004) — the 512 KB RWX region in the WMI host suggests WMI-based execution/persistence; review WMI event subscriptions and `__EventFilter / CommandLineEventConsumer` bindings on the source system.
3. [P2] Triage the zeroed RWX VADs in `explorer.exe` and `chrome.exe` (F-NOTCH-002/003) to confirm whether they are stale injection scaffolding or benign JIT/allocator pages.
4. [P3] Acquire on-disk and longitudinal artefacts (event logs, MFT, registry, network telemetry) to recover the initial-access vector and dwell time, which a single RAM snapshot cannot establish.

► Appendix — methodology, tool versions, chain of custody